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Quantum Simulation of a Two-Spin System using Qiskit

Introduction

Quantum computing is a rapidly growing field that uses the principles of quantum mechanics to process information. Unlike classical computers, which use bits represented by either 0 or 1, quantum computers use quantum bits, or qubits, that can exist in multiple states simultaneously.

One of the most promising applications of quantum computing is the simulation of quantum systems. Classical computers often struggle to simulate large quantum systems because the amount of information required grows exponentially with the number of particles. Quantum computers, however, naturally follow the laws of quantum mechanics and can therefore represent these systems more efficiently.

In this project, a simple two-spin quantum system is simulated using Qiskit. The project demonstrates how quantum computers can prepare entangled states, perform measurements, and estimate physical quantities such as energy.

Quantum Spins and Qubits

A spin is a fundamental quantum property of particles such as electrons. For simplicity, a spin can be imagined as a tiny magnet that can point either upward or downward.

In quantum computing, these two possible orientations are represented using qubit states:

- Spin Up → $|0\rangle$
- Spin Down → $|1\rangle$

Unlike classical systems, a quantum spin is not restricted to only one of these states. It can exist in a superposition, meaning it can be partially in both states at the same time until a measurement is performed.

This ability to exist in multiple states simultaneously is one of the key features that makes quantum computing powerful.

Quantum Entanglement

Another unique property of quantum mechanics is entanglement.

When two qubits become entangled, their states become strongly correlated. Measuring one qubit immediately provides information about the other, regardless of how far apart they are.

To demonstrate entanglement, this project prepares the Bell state:

$|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$

This state is known as a maximally entangled state. When measured, the two qubits always produce opposite outcomes:

- If the first qubit is measured as 0, the second will be 1.
- If the first qubit is measured as 1, the second will be 0.

This perfect anti-correlation is a clear signature of quantum entanglement.

Bell State Preparation

The quantum circuit begins with two qubits initialized in the state:

$|00\rangle$

A Hadamard gate is first applied to the first qubit. This creates a superposition, allowing the qubit to exist in both $|0\rangle$ and $|1\rangle$ simultaneously.

Next, a Controlled-NOT (CNOT) gate is applied. This operation entangles the two qubits and produces the Bell state:

$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$

Finally, additional X and Z gates are applied to transform the state into:

$|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$

This Bell state is used throughout the project for both measurement and energy estimation experiments.

Measurement of the Quantum State

After preparing the Bell state, both qubits are measured.

From theory, only two outcomes are expected:

- 01
- 10

Each outcome should occur approximately 50% of the time.

The outcomes 00 and 11 should ideally never appear. However, when running on real quantum hardware, small numbers of these outcomes may occur because of noise, gate imperfections, and measurement errors.

The measurement results provide evidence that the Bell state was successfully prepared.

Quantum Simulation of a Two-Spin System

The main objective of this project is to simulate a simple physical system consisting of two interacting quantum spins.

In many physical materials, neighboring spins interact with one another and are influenced by external magnetic fields. Understanding the behavior of such systems is an important problem in condensed matter physics, material science, and quantum chemistry.

Quantum computers are particularly well suited for studying these systems because they naturally obey the same quantum mechanical rules as the systems being simulated.

Hamiltonian and Energy

The energy of a quantum system is described by a mathematical operator known as the Hamiltonian.

For this project, the Hamiltonian is defined as:

$H = J(ZZ) + hx(XI + IX)$

where:

- J represents the interaction strength between the two spins.
- hx represents the strength of an external transverse magnetic field.

The first term describes how the two spins interact with each other, while the second term represents the influence of the magnetic field that can flip the spins and create quantum superposition.

By evaluating this Hamiltonian, the energy of the prepared quantum state can be determined.

Qiskit Primitives

Two important Qiskit Runtime primitives are used in this project: Sampler and Estimator.

Sampler

The Sampler primitive is used to determine the probability distribution of measurement outcomes.

It answers the question:

"What states are observed when the circuit is measured?"

The output is a collection of measurement counts that can be visualized using a histogram.

Estimator

The Estimator primitive is used to calculate expectation values of observables.

Instead of producing measurement counts, it answers the question:

"What is the value of a physical quantity for the prepared quantum state?"

In this project, Estimator is used to calculate the energy of the Bell state according to the Hamiltonian.

Circuit Optimization and Transpilation

Before a quantum circuit can run on real hardware, it must be translated into instructions supported by the target quantum processor.

This process is called transpilation.

During transpilation:

- Logical qubits are mapped to physical qubits.

- Unsupported gates are decomposed into native hardware gates.
- Circuit depth is optimized.
- Hardware constraints are taken into account.

Qiskit automatically performs these tasks using a preset pass manager.

Real Quantum Hardware and Simulation

The experiments in this project are executed using both IBM Quantum hardware and the Aer Simulator.

Running on real hardware demonstrates practical quantum computation and provides insight into real-world device behavior. However, execution may involve queue delays and hardware noise.

The Aer Simulator provides a faster alternative that can reproduce theoretical results without waiting for access to a physical quantum processor.

Comparing simulator results with hardware results helps illustrate the effects of noise and imperfections in current quantum devices.

Results and Discussion

The measurement results obtained from the Bell state show strong anti-correlation between the two qubits, confirming the presence of quantum entanglement.

The energy calculated using the Estimator primitive is close to the expected theoretical value for the chosen Hamiltonian parameters. Minor deviations may occur on real hardware due to noise and gate errors.

These results demonstrate how quantum computers can be used not only to prepare and measure quantum states but also to study the physical properties of quantum systems.

Conclusion

This project provides a practical introduction to quantum computing and quantum simulation using Qiskit and IBM Quantum services.

Through the preparation of a Bell state, measurement using Sampler, and energy estimation using Estimator, the project demonstrates several fundamental concepts including superposition, entanglement, quantum measurement, Hamiltonians, and quantum simulation.

Although the example involves only two qubits, the same principles can be extended to larger and more complex quantum systems. As quantum hardware continues to improve, quantum simulation is expected to become one of the most impactful real-world applications of quantum computing.